Marking Schemes

This document was prepared for markers' reference. It should not be regarded as a set of model answers. Candidates and teachers who were not involved in the marking process are advised to interpret its contents with care.

General Marking Instructions

1. It is very important that all markers should adhere as closely as possible to the marking scheme. In many cases, however, candidates may have obtained a correct answer by an alternative method not specified in the marking scheme. In general, a correct answer merits ***the answer mark*** allocated to that part, unless a particular method has been specified in the question.

   In the marking scheme, alternative answers and marking guidelines are in $\boxed{\text{rectangles}}$.

2. In the marking scheme, answer marks or 'A' marks are awarded for a correct numerical answer with a unit. If the answer should be in km, then cm and m are considered to be wrong units.

3. In a question consisting of several parts each depending on the previous parts, method marks or 'M' marks are awarded to steps/methods or substitutions correctly deduced from previous answers.

4. In cases where a candidate answers more questions than required, the answers to all questions should be marked. However, the excess answer(s) receiving the lowest score(s) will be disregarded in the calculation of the final mark.

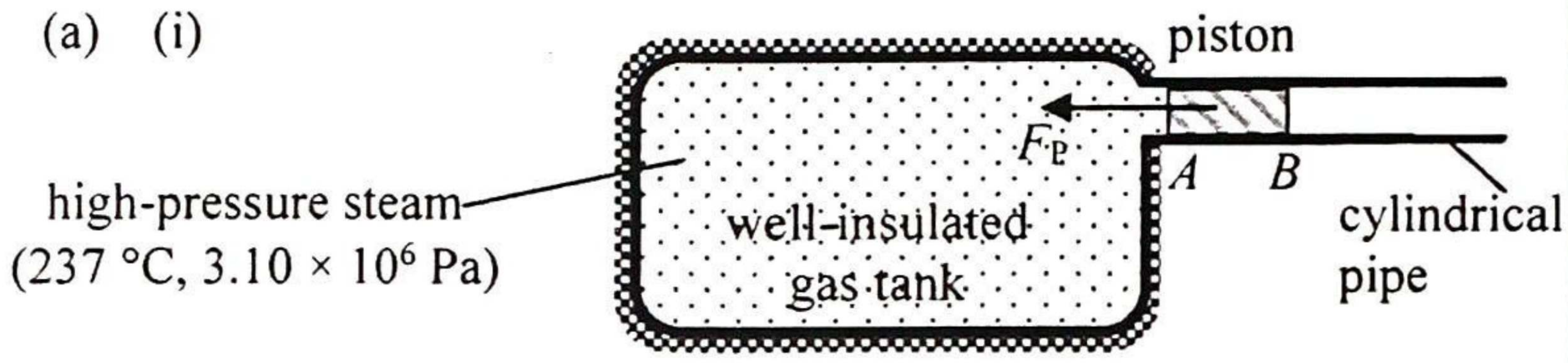
Paper 1 Section A

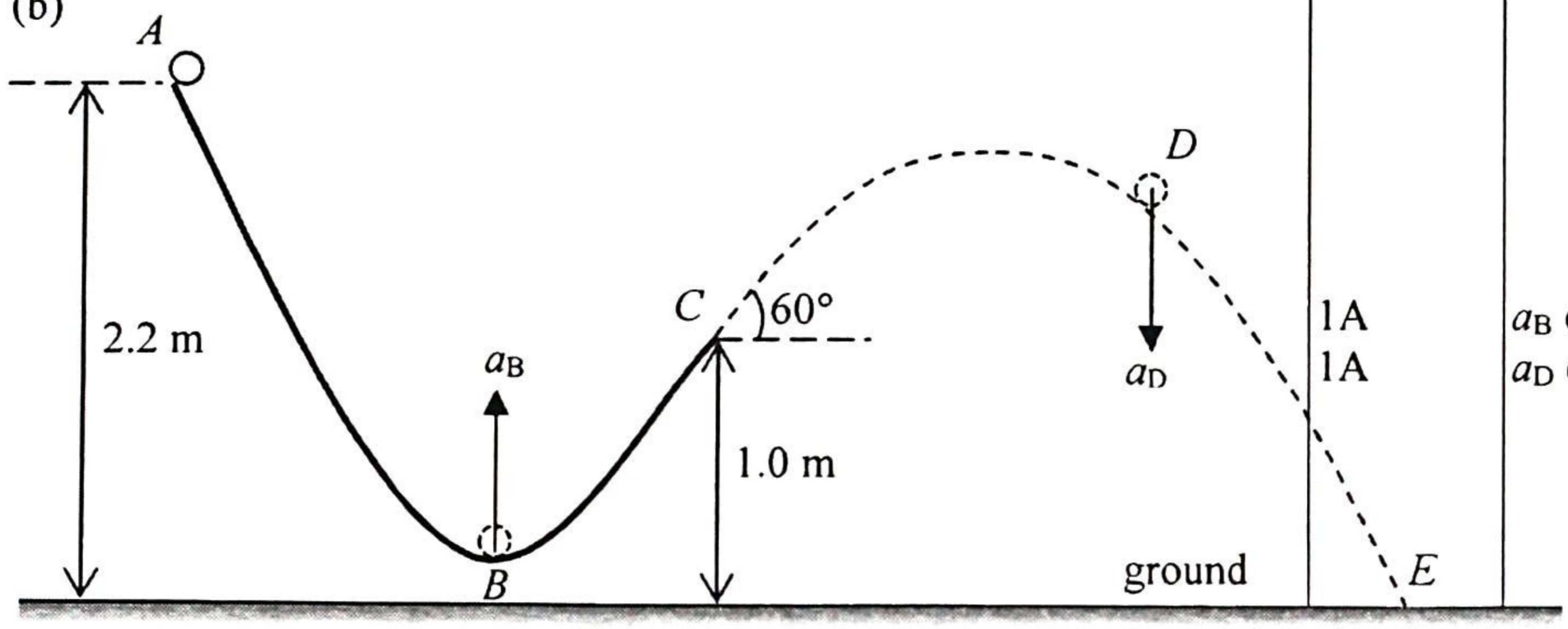
Question No.	Key	Question No.	Key
1.	D (56)	26.	A (42)
2.	C (65)	27.	B (38)
3.	D (52)	28.	D (32)
4.	A (64)	29.	B (56)
5.	C (52)	30.	D (63)
6.	B (66)	31.	B (55)
7.	D (29)	32.	D (49)
8.	C (81)	33.	C (38)
9.	A (59)		
10.	C (60)		
11.	D (59)		
12.	C (45)		
13.	C (67)		
14.	A (70)		
15.	C (52)		
16.	A (45)		
17.	D (41)		
18.	A (45)		
19.	B (49)		
20.	B (31)		
21.	A (75)		
22.	B (62)		
23.	B (32)		
24.	D (52)		
25.	A (71)		

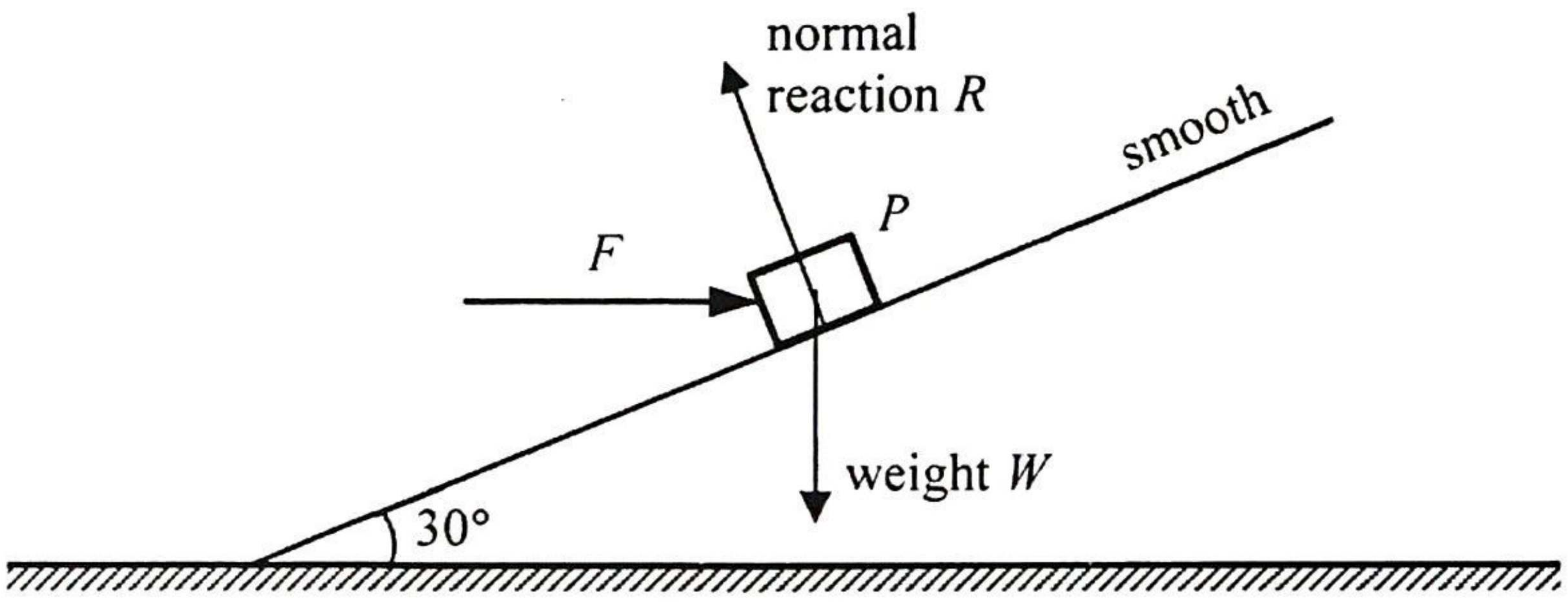
Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

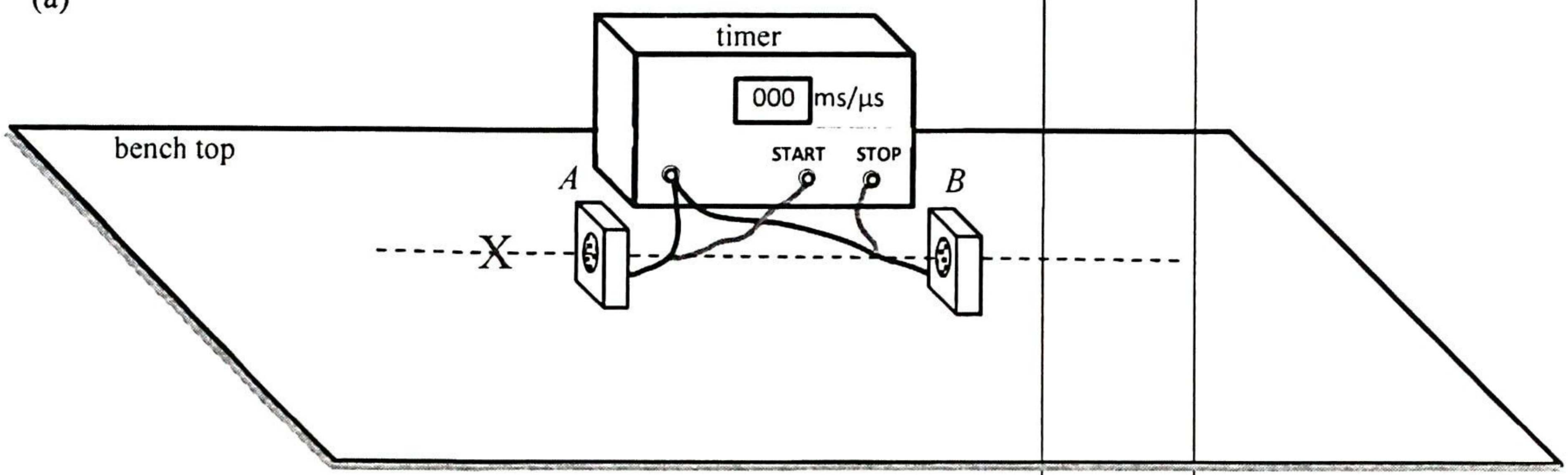
Paper 1 Section B

Solution	Marks	Remarks
1. (a) Let T be the final temperature of the mixture. $(5 \times 0.02) \times 3300 \times (T - 4) = 0.60 \times 4200 \times (96 - T)$ $T = 85.347368\ °C \approx 85.3\ °C$	1M 1A	 Accept 85.0 °C to 85.4 °C
	2	
(b) (i) To compensate for / balance the heat loss (of the container with soup) to the surroundings.	1A	
	1	
(ii) $P \times 10 \times 60 = 2000 \times 9 + 16 \times 4200 \times 9$ $P = 1038\ W \approx 1040\ W$	1M+1M 1A	Assume : Power of the heater = Rate of heat loss (of the container with soup) to the surroundings
	3	
(iii) Smaller than 9 °C. As the temperature of (the container with) soup drops, the rate of heat loss becomes lower due to a smaller temperature difference (with respect to the surroundings).	1A 1A	
	2	

Solution	Marks	Remarks
2. (a) (i) 	1A	F_P correctly marked
	1	
(ii) $F_P = (3.10\times10^6 - 1.0\times10^5)\times0.67$	1M	
$= 2010000\text{ N} = 2.01\times10^6\text{ N}$	1A	Accept 2.00×10^6 N to 2.01×10^6 N
	2	
(iii) $pV = nRT \Rightarrow V = \dfrac{nRT}{p}$		
$V = \dfrac{(570/0.018)(8.31)(237+273)}{3.10\times10^6}$	1M+1M	
$= 43.292419\text{ m}^3 \approx 43.3\text{ m}^3$	1A	Accept 43.0 m^3 to 43.3 m^3
	3	
(b) (i) Work done = K.E. gained		
$= \frac{1}{2}(2.6\times10^4)\,54^2$	1M	
$= 3.7908\times10^7\text{ J} \approx 37.9\text{ MJ}$	1A	Accept 37.9 MJ to 38.0 MJ
	2	
(ii) Average acceleration $a = \dfrac{v-u}{t} = \dfrac{54-0}{1.5}$	1M	
$= 36\text{ m s}^{-2}$	1A	
	2	
(iii) Acceleration is decreasing (i.e. maximum at first).	1A	
(According to kinetic theory,) once the steam expands, i.e. volume increases, its pressure decreases,	1A	
thus the (pressure) force acting on the piston at A decreases, hence a smaller acceleration.	1A	
	3	
3. (a) With superconductor / extremely low or nearly zero resistance,	1A	
a much larger current flows in the coils producing a stronger magnetic field (with less heat loss due to the current flow).	1A	
	2	
(b) P and Q : N (north pole)	1A	
Repulsive force between like poles	1A	
	2	
(c) (i) As the train is not in contact with the rail, there is no friction / interaction between the train and the rail, thus smoother due to less vibration.	1A	
(ii) There is no friction / interaction between the train and the rail, and the propulsion of the train only needs to work against air resistance, therefore much faster.	1A	
	2	

Solution	Marks	Remarks
4. (a) $v_B > v_C > v_D > v_A$	1A	
	1	
(b) 	1A 1A	a_B correctly marked a_D correctly marked
	2	
(c) (i) Gravitational potential energy is changed to kinetic energy from A to B, and	1A	
some kinetic energy is changed back to gravitational potential energy from B to C.	1A	
	2	
(ii) $\frac{1}{2}mv^2 = mgh$		
$v^2 = 2(9.81)(2.2 - 1.0)$	1M	For $g = 10\text{ m s}^{-2}$,
$v = 4.852216\text{ m s}^{-1} \approx 4.85\text{ m s}^{-1}$	1A	$v = 4.89898\text{ m s}^{-1} \approx 4.90\text{ m s}^{-1}$ Accept 4.85 m s^{-1} to 4.90 m s^{-1}
	2	
(iii) Horizontal speed (at C) $= 4.85 \times \cos 60° = 2.426108\text{ m s}^{-1} \approx 2.43\text{ m s}^{-1}$	1M	For $g = 10\text{ m s}^{-2}$, horizontal speed $= 2.44949\text{ m s}^{-1} \approx 2.45\text{ m s}^{-1}$
Distance $= 2.43 \times t = 2.55\text{ m}$	1M	
$t = 1.051066\text{ s} \approx 1.05\text{ s}$	1A	Accept 1.04 s to 1.10 s
Or Vertical speed (at C) $= 4.85 \times \sin 60° = 4.202142\text{ m s}^{-1} \approx 4.20\text{ m s}^{-1}$	1M	For $g = 10\text{ m s}^{-2}$, vertical speed $\approx 4.24\text{ m s}^{-1}$
$-1.0 = 4.20 \times t + \frac{1}{2}(-9.81)t^2$	1M	and $t = 1.041033\text{ s} \approx 1.04\text{ s}$
$t = 1.051066\text{ s} \approx 1.05\text{ s}$	1A	
	3	

Solution	Marks	Remarks
5. (a) (i) 	1A 1A	Forces R(or N) and W(or Mg) correctly indicated and labelled.
	2	
(ii) $R\cos 30° = W = Mg$ and $R\sin 30° = F$	1M	$M = 10$ kg; $Mg = 98.1$ N
$F = Mg \times \tan 30° = 56.63806\text{ N} \approx 56.6\text{ N}$	1A	For $g = 10\text{ m s}^{-2}$, $57.735027\text{ N} \approx 57.7\text{ N}$
$N = R = \dfrac{Mg}{\cos 30°} = 113.27612\text{ N} \approx 113\text{ N}$	1A	$115.470054\text{ N} \approx 115\text{ N}$
Or $R = W\cos 30° + F\sin 30°$ and $W\sin 30° = F\cos 30°$	1M	
	3	
(b) (i) $g\sin\theta = 9.81\sin 30° = 4.905\text{ m s}^{-2} \approx 4.91\text{ m s}^{-2}$	1A	5 m s^{-2} for $g = 10\text{ m s}^{-2}$
	1	
(ii) Decrease	1A	
as the component of F perpendicular to the incline no longer acts on the block / incline	1A	
Or as F is removed, the force pressing the block / incline would decrease (only the weight's component left)	1A	
	2	

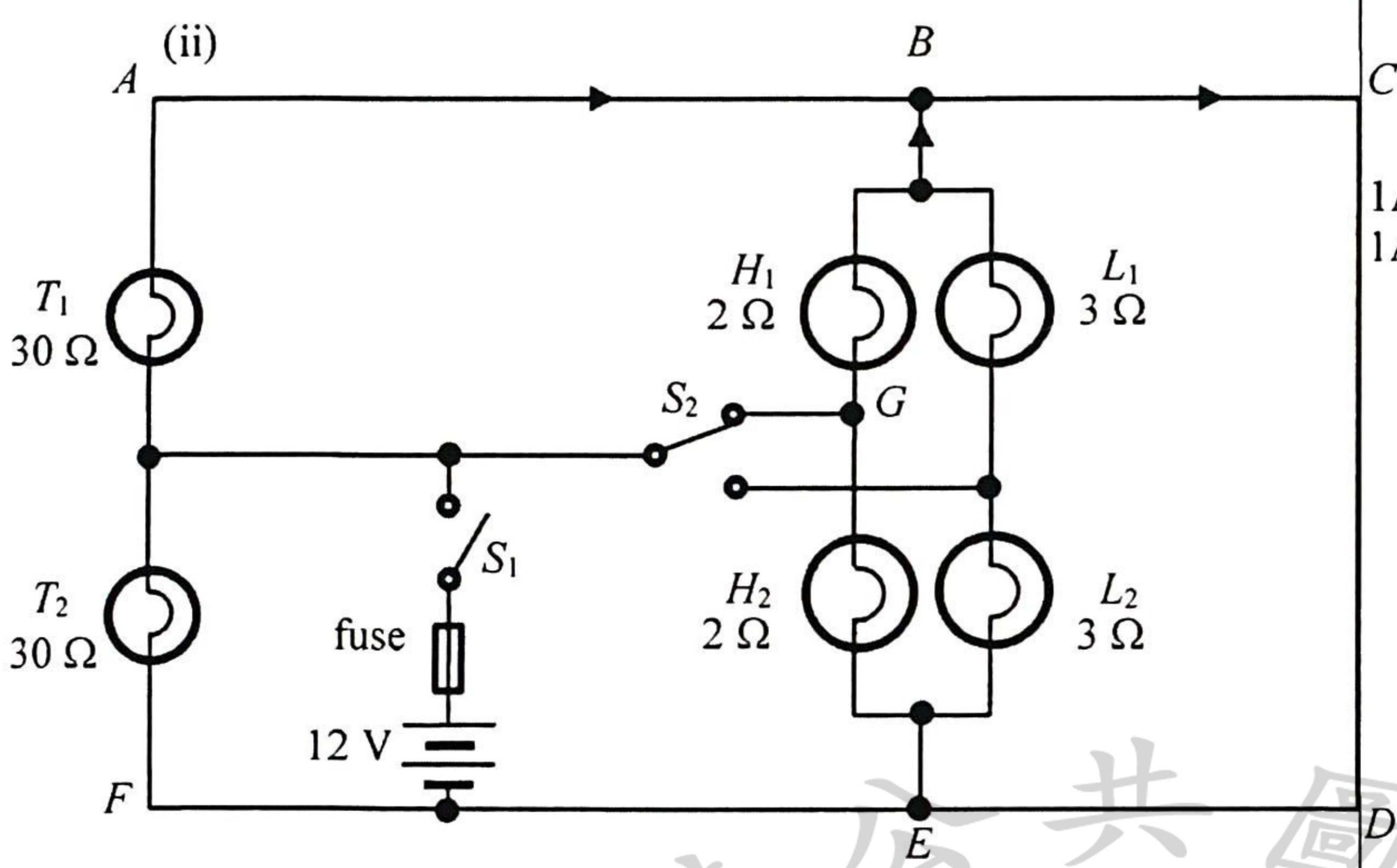
<table>
<tr><th>Solution</th><th>Marks</th><th>Remarks</th></tr>
<tr><td>6. (a)

</td><td></td><td></td></tr>
<tr><td>Correct location of 'X' (close to axis and on the side of the START microphone A).</td><td>1A</td><td></td></tr>
<tr><td>Use a metre rule
to measure the distance / separation D between the two microphones.
Record the timer reading, which is the time interval Δt for the sound to travel from one microphone (START) to the other microphone (STOP), i.e. A to B.</td><td>1A

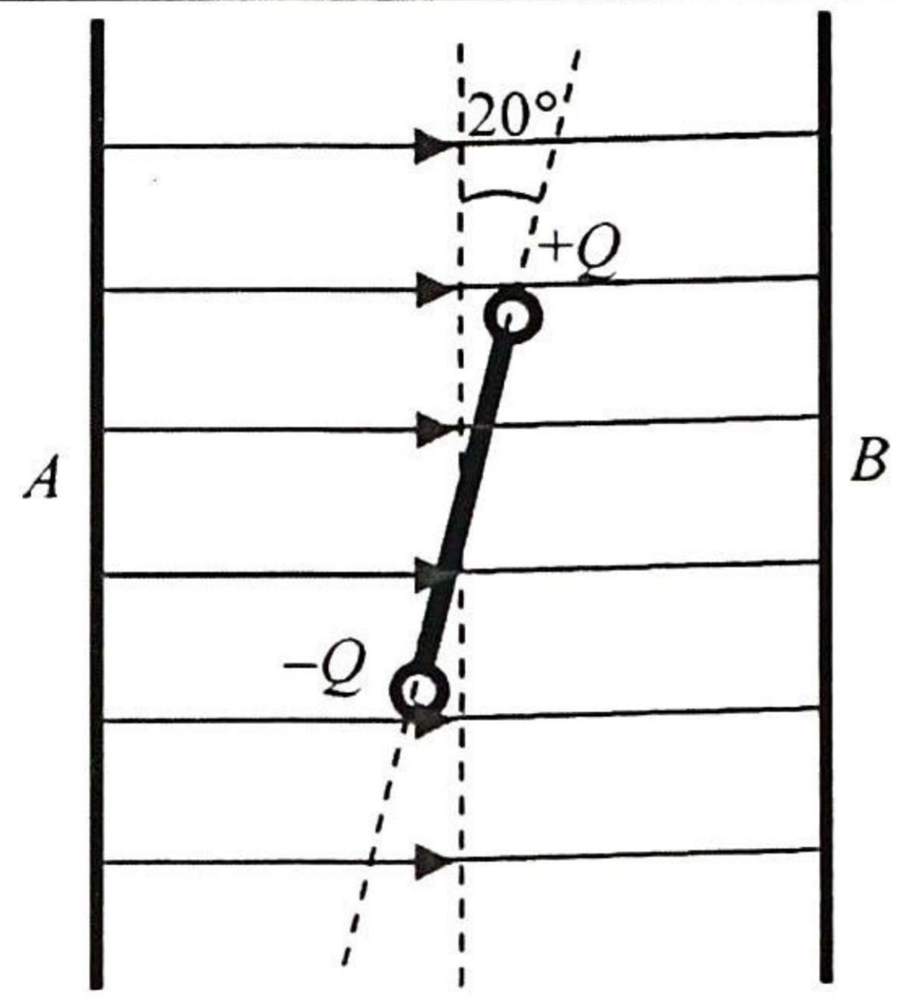
1A</td><td>Any ONE of the measurements</td></tr>
<tr><td></td><td>3</td><td></td></tr>
<tr><td>(b) (i) Discard 539 μs, $\Delta t = \dfrac{801+838+821}{3} = 820\ \mu s$
The speed of sound in air $v = \dfrac{D}{\Delta t}$
$v = \dfrac{0.280}{820 \times 10^{-6}}$
$= 341.463415\ m\ s^{-1} \approx 341\ m\ s^{-1}$</td><td>1M

1A</td><td>

Accept 340 m s^{-1} to 342 m s^{-1}</td></tr>
<tr><td>(ii) Increase separation D between the microphones.</td><td>1A</td><td></td></tr>
<tr><td></td><td>3</td><td></td></tr>
</table>

Solution	Marks	Remarks
7. (a) (i) Angle of incidence at A, $i_A = 90° - 30° = 60°$	1A	
	1	
(ii) $n_g \sin c = n_c \sin 90°$	1M	
$\Rightarrow \frac{n_g}{n_c} = \frac{1}{\sin c} \geq \frac{1}{\sin 60°} = 1.1547005 \approx 1.15$	1A	
	2	
(iii) Total internal reflection.	1A	
For light rays from O with $\theta > 30°$, they will fail to undergo total internal reflection.	1A	
	2	
(b) (i) Some light / energy (of the narrow light pulse) taking the shortest path (i.e. OD) arrives first while the rest of the	1A	1A for 'taking different paths'
energy taking longer paths arrives later. Therefore the pulse is broader and the height of the pulse is lower (smaller intensity).	1A	1A for 'different arrival time'
	2	
(ii) The refractive index n_c of cladding should be increased.	1A	Note: More light rays will escape from the optical fibre for a larger n_c.
As $\frac{n_g}{n_c} = \frac{1}{\sin c}$, by making c / $\sin c$ larger, only those light rays close to the axis / within a smaller θ are totally internal reflected, (i.e. $\frac{n_g}{n_c}$ getting closer to 1).	1A	
	2	

Solution	Marks	Remarks
8. (a) Because L_1 and L_2 (connected in series) are shorted by $BCDE$.	1A	Accept : p.d. across each low-beam headlight (L_1 or L_2) is 0.
	1	
(b) (i) p.d. across T_2 = 12 V	1A	
	1	
(ii) 	1A 1A	Any two currents correctly marked. All currents correct.
largest current in branch BC	1A	
	3	
(c) Power supplied $P = 2\times\left(\frac{12^2}{30}+\frac{12^2}{2}\right)$	1M	
$= 153.6\text{ W} \approx 154\text{ W}$	1A	
$\frac{V^2}{R_{eq}} = \frac{12^2}{R_{eq}} = 153.6$ (or $\frac{1}{R_{eq}} = \frac{1}{30}+\frac{1}{30}+\frac{1}{2}+\frac{1}{2}$)	1M	
$R_{eq} = 0.9375\ \Omega \approx 0.938\ \Omega \approx 1\ \Omega$	1A	
	4	
(d) maximum current $I_{max} = \frac{V}{R_{eq}} = \frac{12}{0.9375} = 12.8$ A (or $I_{max} = \frac{P}{V} = \frac{153.6}{12} = 12.8$ A)	1M	
Since the rating is slightly higher than the max. current (when high-beam headlights and taillights are switched on), this fuse is suitable.	1A	
	2	

Solution	Marks	Remarks
9. (a) Top view 	1A 1A 2	Correct direction (From A to B) Perpendicular to plates, parallel and evenly spaced
(b) (i) $F \times d = (2.0 \times 10^{-5})(0.05 \cos 20°)$ $= 9.396926 \times 10^{-7}$ N m $\approx 9.40 \times 10^{-7}$ N m	1M 1A 2	
(ii) $E = \frac{V}{d}$ $= \frac{5.0 \times 10^3}{0.1}$ $= 50\ 000$ V m^{-1} or N C^{-1} = 50 kV m^{-1} or kN C^{-1}	1M 1A 2	
(iii) $E = \frac{F}{Q}$ $Q = \frac{F}{E} = \frac{2.0 \times 10^{-5}}{5.0 \times 10^{4}}$ $= 4.0 \times 10^{-10}$ C	1M 1A 2	
10. (a) proton / ${}^{1}_{1}$H / p / hydrogen nucleus	1A 1	
(b) Change in mass $\Delta m = (16.9947 + 1.0073) - (13.9993 + 4.0015)$ $= 0.0012$ u Energy required $= 0.0012 \times 931$ $= 1.1172$ (MeV) ≈ 1.12 (MeV)	1M 1A 2	Accept 1.10 (MeV) to 1.12 (MeV)
(c) By the conservation of momentum, as before the reaction occurs the α particle has momentum, the total momentum of the products (= momentum of the α particle) must also be non-zero, i.e. the total K.E. of the products must greater than zero, thus the α particle should have a larger K.E. (than that found in (b)).	1A 1A 2	[The kinetic energy of the α particle = energy calculated in (b) + total K.E. of the products]

Paper 2

Section A : Astronomy and Space Science

1. D (60%)	2. D (24%)	3. C (63%)	4. B (44%)
5. C (55%)	6. B (57%)	7. C (45%)	8. A (52%)

Solution	Marks	Remarks
1. (a) Distance 50 kpc = 50000 × 3.26 ly = 163000 ly Thus the explosion of the star took place 163000 years ago.	1A	Note: 2020 – 1987 = 33 years can be ignored. Accept: (163000 ~ 164000) years
	1	
(b) SN 1987A at 50 kpc, being much farther away than 10 pc, would be much brighter (than that corresponding to +2.9) if it were placed at 10 pc.	1A	
Hence (the numerical value of) its absolute magnitude is much smaller than +2.9 / the apparent magnitude.	1A	
	2	
(c) (i) Take L_S, R_S and T_S to be the luminosity, radius and surface temperature of the Sun respectively, whereas let L_X, R_X and T_X be those of X respectively. By Stefan's law, $L_S = \sigma(4\pi R_S^2)T_S^4$ and $L_X = \sigma(4\pi R_X^2)T_X^4$		
Therefore, $\frac{L_X}{L_S} = \left[\frac{R_X}{R_S}\right]^2\left[\frac{T_X}{T_S}\right]^4$	1M	
$40000 = \left[\frac{R_X}{R_S}\right]^2 [3.1]^4$	1M	
$\frac{R_X}{R_S} = 20.81165 \approx 20.8$		
	2	
(ii) Region A.	1A	
Not a 'red giant' as the temperature of star X is (much) higher than that of the Sun.	1A	Note: Star X is in fact a blue supergiant.
Or Red giants are in region B	1A	
	2	
(d) Q: L_0 ; R: L_1	1A	
According to Doppler effect, gas at R receding from the observer gives rise to red shift (vice versa for P),	1A	
while there is no velocity component for gas at Q (and S) towards / away from the observer, no Doppler / blue / red shift.	1A	
	3	

Section B : Atomic World

1. C (29%)	2. C (75%)	3. B (27%)	4. D (23%)
5. A (40%)	6. A (42%)	7. B (61%)	8. D (32%)

Solution	Marks	Remarks
2. (a) (i) ultra-violet (UV)	1A	
	1	
(ii) Light (energy) is (transferred to electrons of the cathode) in packets or (whole number of) quanta (i.e. quantized) called photons.	1A	
	1	
(b) (i) Microammeter reading remains zero,	1A	
energy E of incident photons remains unchanged, although increase in intensity would cause more photons incident but there is no effect on maximum K.E. of the electrons emitted or on photoemission.	1A	
	2	
(ii) Energy of photon $= \dfrac{hc}{\lambda}$ $= \dfrac{(6.63\times10^{-34})(3\times10^{8})}{300\times10^{-9}}$	1M	
$= 6.63\times10^{-19}$ J		
$= 4.14375$ (eV) ≈ 4.14 (eV)		Accept 4.10 (eV) to 4.14 (eV)
Work function $\Phi = 4.14 - 1.7$	1M	
$= 2.44375$ (eV) ≈ 2.44 (eV)	1A	Accept 2.40 (eV) to 2.44 (eV)
	3	
(c) (i) No. of photoelectrons reaching A in 1 s $= \dfrac{0.4\times10^{-6}}{1.6\times10^{-19}} = 2.5\times10^{12}$	1A	
	1	
(ii) $1.7 - 0.8 = 0.9$ (eV) <u>or</u> $4.14 - 2.4 - 0.8 = 0.94$ (eV)	1A	
Electrons inside cathode C (not on surface) need an amount of energy more than the work function to escape / be emitted from C.	1A	
	2	

Section C : Energy and Use of Energy

1. B (66%)	2. A (83%)	3. D (73%)	4. C (55%)
5. A (65%)	6. B (24%)	7. D (48%)	8. A (85%)

Solution	Marks	Remarks
3. (a) The fission products / nuclides are with higher binding energy per nucleon than uranium-235.	1A	
Therefore, energy is released in fission and the resulting nuclides are more stable.	1A	
	2	
(b) (i) It represents the energy required to separate all nucleons (protons and neutrons) of uranium-235 to infinity / far apart / separate completely.	1A	Accept: Energy released when protons and neutrons (nucleons) form a single nucleus.
	1	
(ii) Binding energy (B.E.) of $^{235}_{92}U$ nucleus = 1783 MeV		
B.E. of $^{144}_{56}Ba$ nucleus = 8.27 × 144 = 1190.88 MeV B.E. of $^{90}_{36}Kr$ nucleus = 8.59 × 90 = 773.1 MeV	1M	
Hence, energy released in fission = (1190.88 + 773.1) – 1783 = 180.98 (MeV) ≈ 181 (MeV)	1A	Accept: 180 (MeV) to 181 (MeV)
	2	
(c) (i) $\dfrac{\text{Total energy released} \times \text{efficiency}}{\text{Power output}}$		
$= \dfrac{(1.30\times10^{30}\times10^{6})(1.6\times10^{-19})(0.4)}{500\times10^{6}}$	1M	
$= 1.664 \times 10^{8}$ s ≈ 5.28254 years ≈ 5.28 years	1A	Accept: 5.27 years to 5.30 years
	2	
(ii) The concentration of uranium-235 nuclei will decrease with time and chain reaction cannot be maintained when the concentration is too low.	1A	
	1	
(d) (i) moderator: to slow down the fast neutrons produced in fission.	1A	
	1	
(ii) control rods: to control the rate of nuclear fission / reaction by absorbing the neutrons <u>or</u> for shutting down the reactor in case of emergency.	1A	
	1	

Section D : Medical Physics

1. D (59%)	2. C (59%)	3. D (59%)	4. A (46%)
5. B (43%)	6. C (76%)	7. B (54%)	8. A (37%)

Solution	Marks	Remarks
4. (a) X-rays are produced when high-speed electrons strike a heavy metal target (like tungsten).	1A	
	1	
(b) (i) density of bone is higher / bone contains element(s) with high atomic number / heavy elements like calcium in bone block X-rays. (Accept other reasonable answers.)	1A	
	1	
(ii) $I = I_0 e^{-\mu_s t_s} = I_0 e^{-0.51\times 5.6}$ $I = I_0 e^{-\mu_b t_b} = I_0 e^{-2.46\times t_b}$ $\therefore 0.51 \times 5.6 = 2.46 \times t_b$	1M	
$t_b = 1.16097561$ cm ≈ 1.2 cm (2 sig. fig.)	1A	Accept 1.16 cm to 1.2 cm
	2	
(iii) Imaging of the breast only involves soft tissues and X-rays of longer wavelength / lower frequency / less penetrating power are needed	1A	
Or giving better contrast of soft tissues / more sensitive to small changes in density of soft tissues.	1A	
For structures with bone, X-rays of shorter wavelength / higher frequency / greater penetrating power are needed.	1A	
	2	
(c) (i) Induce (cause) cancer / heritable mutations / genetically associated diseases. (Accept other reasonable answers.)	1A	
	1	
(ii) A CT scan delivers a 2-D cross-sectional image of the body part by means of multiple exposure (using a rotating X-ray source(s)), equivalent dose is larger due to a relatively longer time of exposure.	1A 1A	
	2	
(iii) cosmic rays / radon gas (from building) / radioactive substances in soil / rock / food or water. (Accept other reasonable answers.)	1A	
	1	

Candidates' Performance

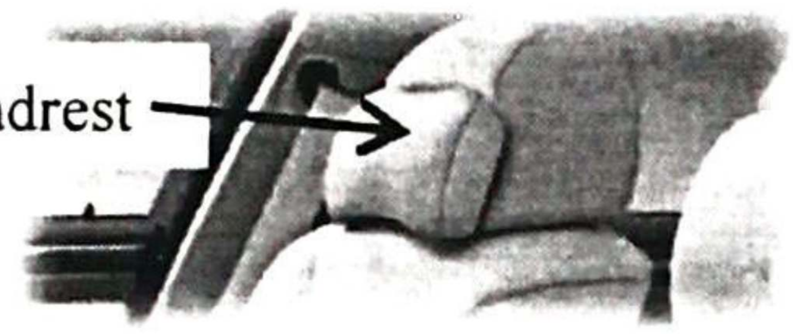
Paper 1

Paper 1 consists of two sections, multiple-choice questions in Section A and conventional questions in Section B. All questions in both sections are compulsory.

Section A (multiple-choice questions)

Section A consisted of 33 multiple-choice questions and the mean score was 18. Items where candidates' performance was typically weaker will be presented below with mean percentage statistics.

5.



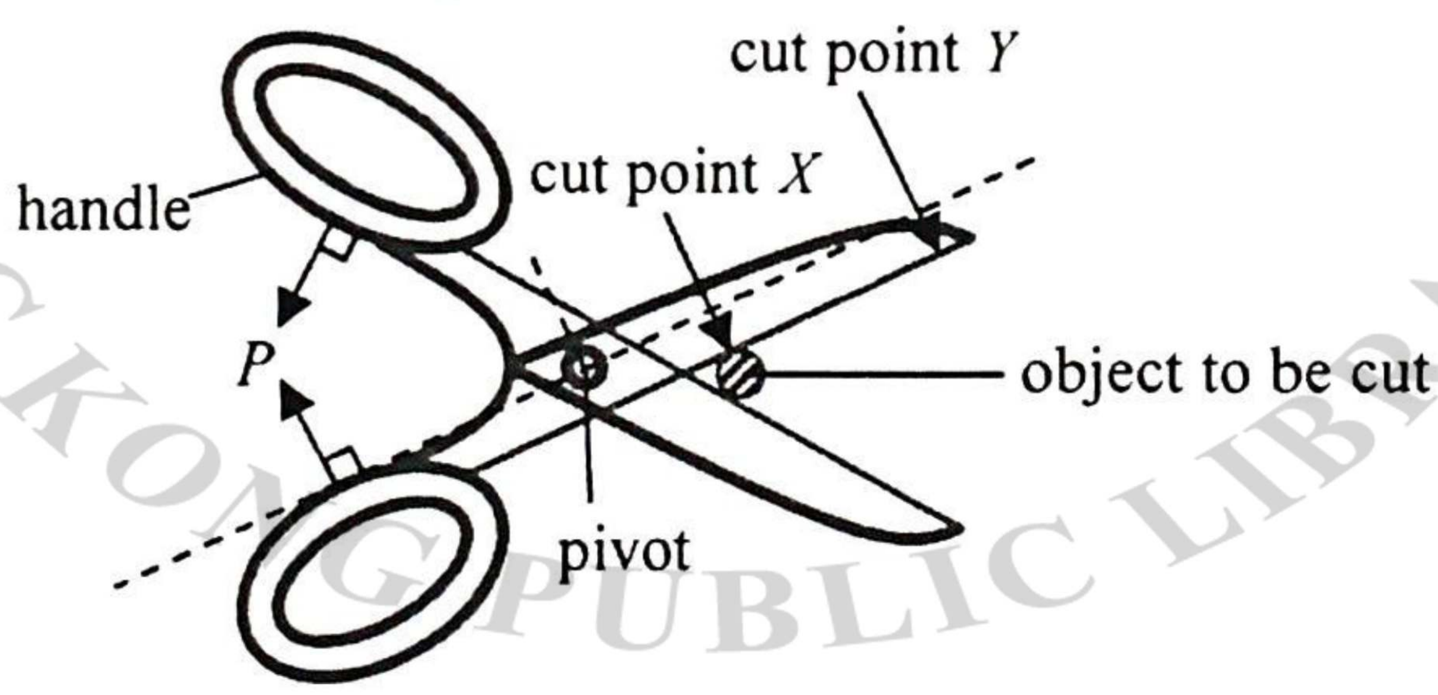
For a car travelling on a highway, which of the following statements about the safety design of the headrest is/are correct ?

(1) As the headrest is soft, it can reduce the force exerted on the passenger's head during impact.
(2) It can minimise injury of the passenger when the car is struck by another one from behind.
(3) It can minimise injury of the passenger when the car brakes suddenly.

A.	(1) only	(25%)
B.	(3) only	(10%)
* C.	(1) and (2) only	(52%)
D.	(2) and (3) only	(13%)

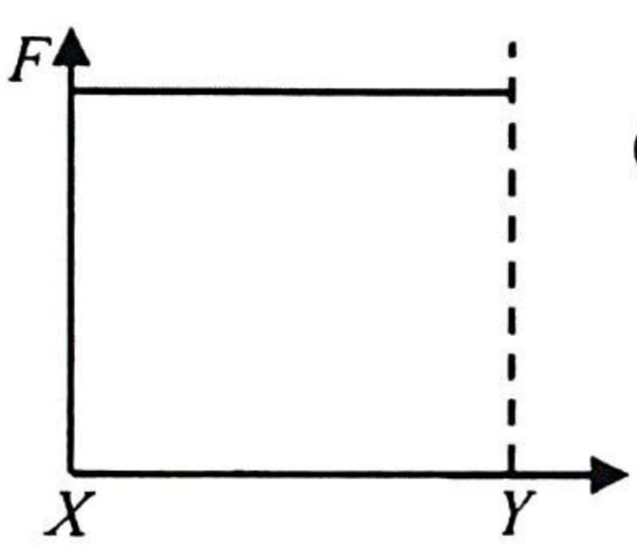
Over one-third of the candidates did not think that statement (2) could be correct.

7. The figure shows that a pair of forces P of constant magnitude is applied at right angles to the handles of a pair of scissors in order to cut an object.



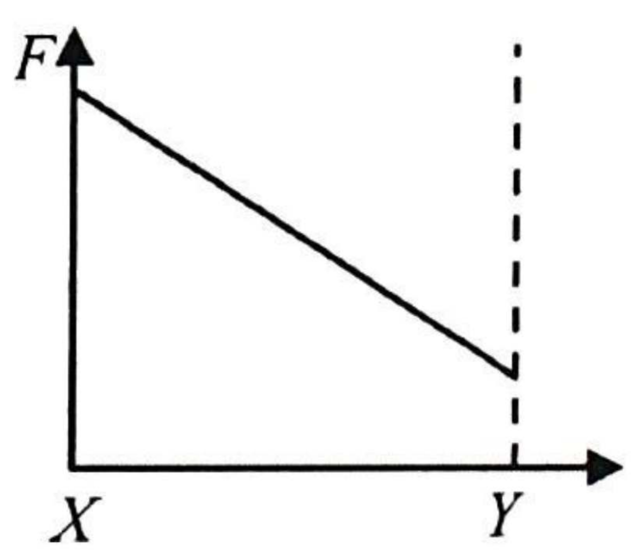
Which graph below best shows the variation of force F produced at cut point from X to Y when the pair of scissors is closed ?

A.



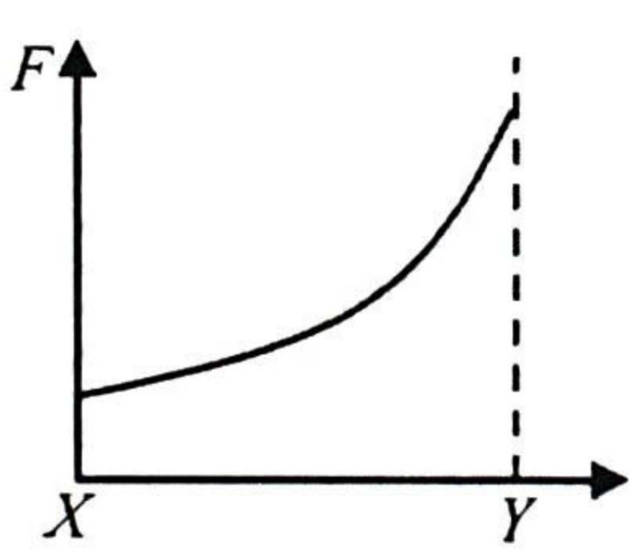
(8%)

B.



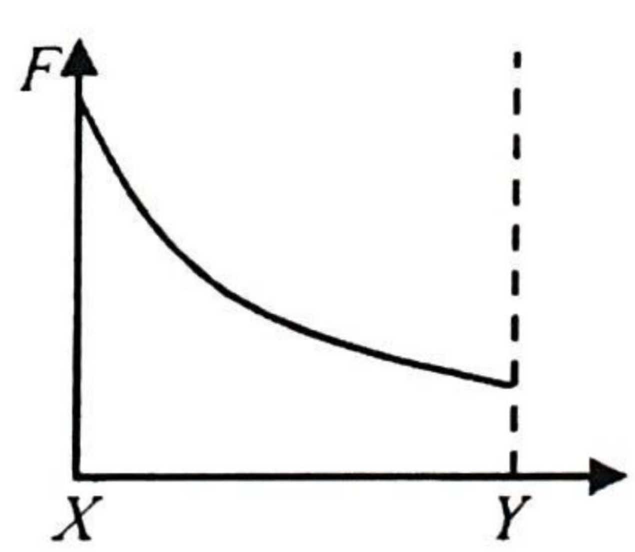
(43%)

C.



(20%)

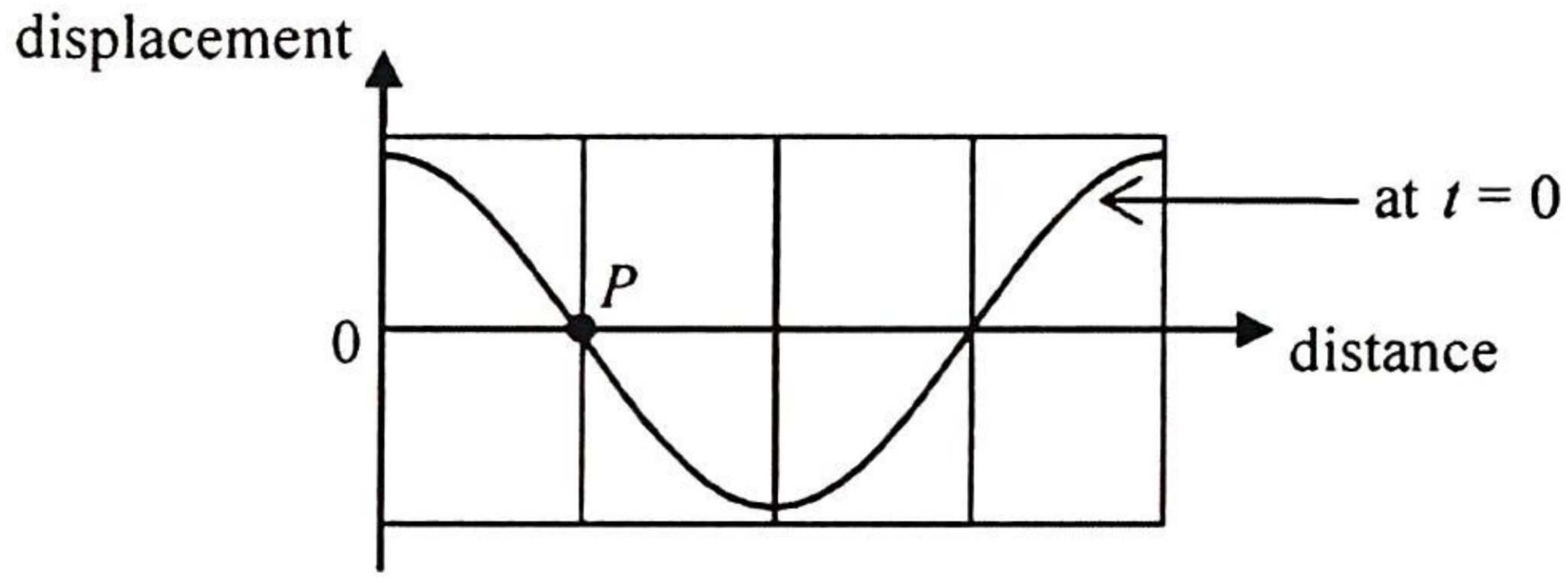
*D.



(29%)

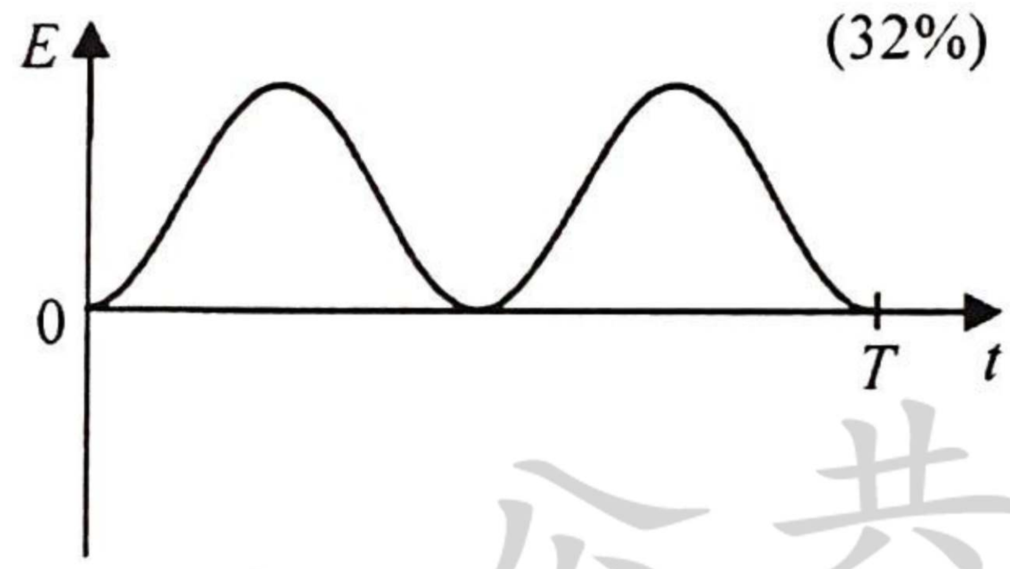
While over 70% of the candidates indicated knowledge that the force is larger at cut point X, less than 30% of them fully understood how the force varies when the pair of scissors is closed.

12. The figure shows part of the displacement-distance graph of a travelling wave of period T at time $t = 0$. P is a particle on the wave.

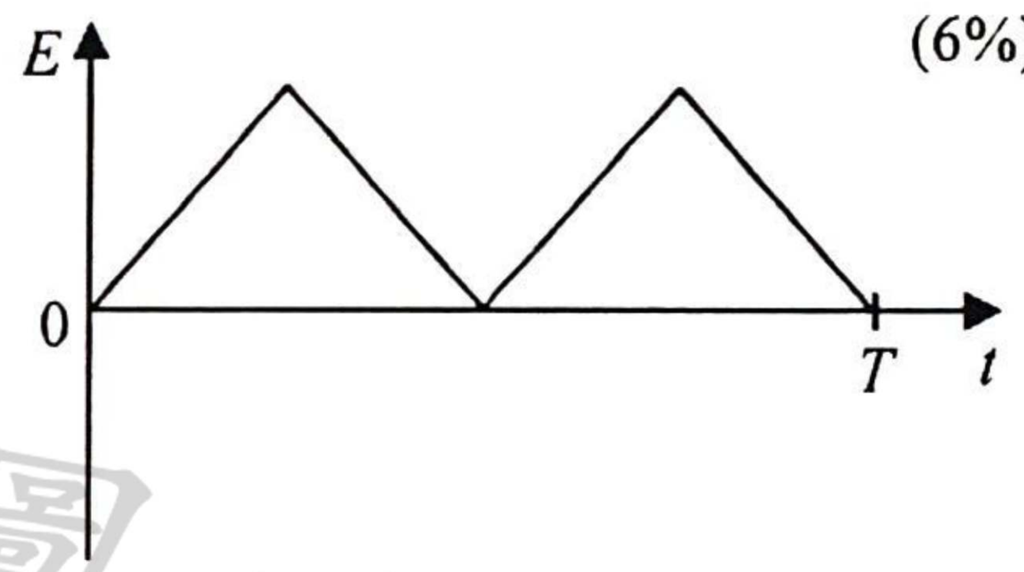


Which graph below correctly shows the variation of the particle's kinetic energy E within a period starting from $t = 0$?

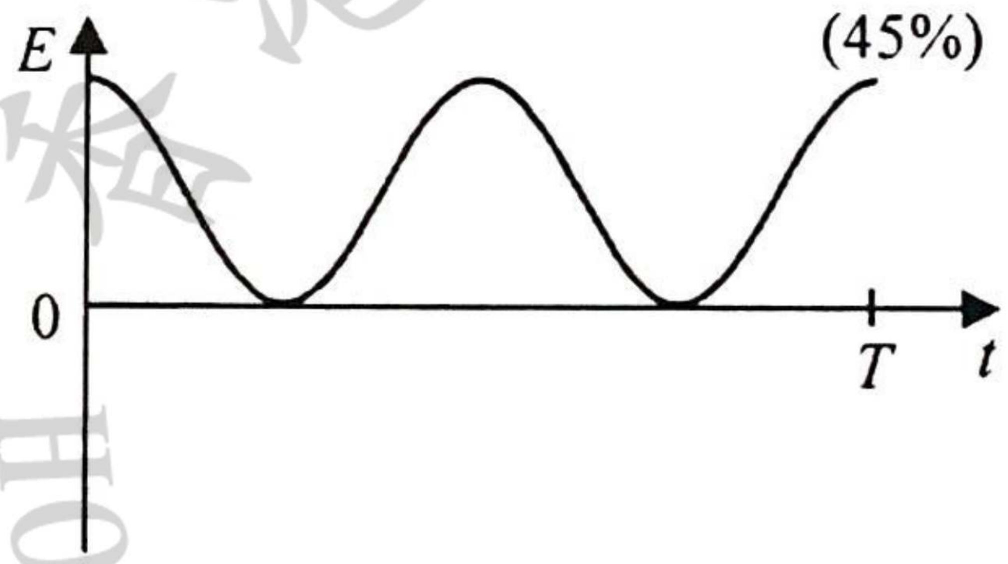
A. (32%)



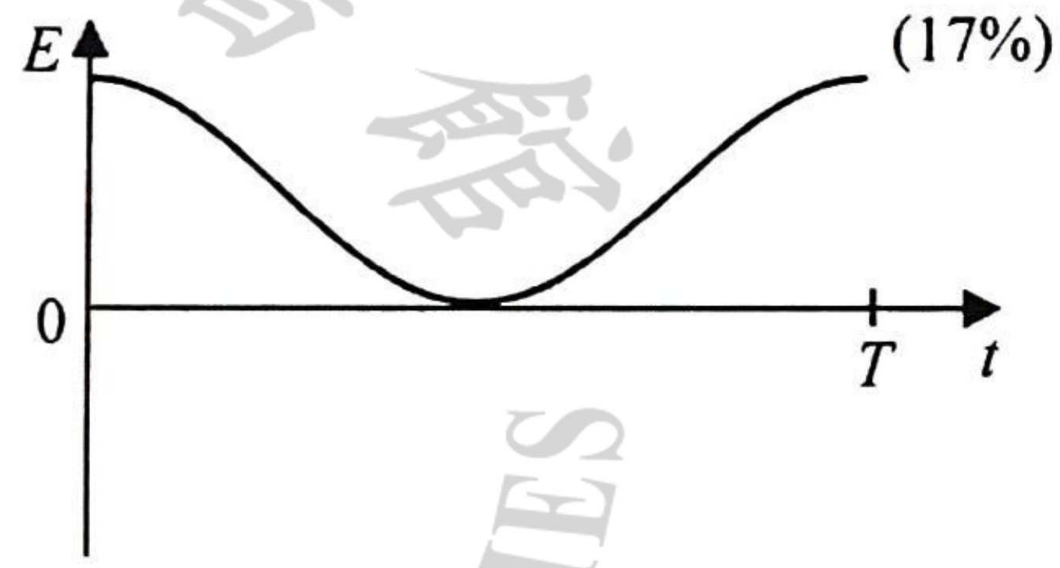
B. (6%)



*C. (45%)



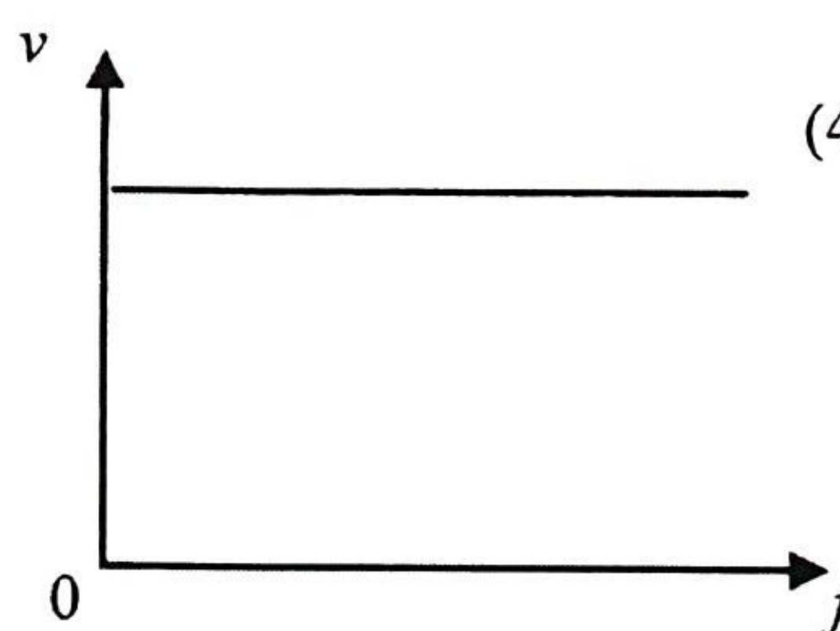
D. (17%)



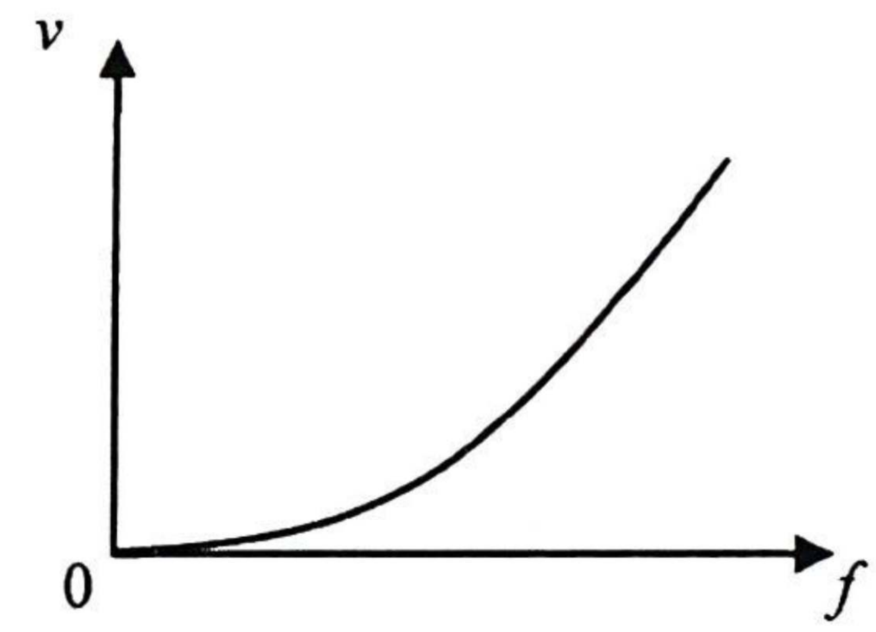
Less than half of the candidates were able to obtain the correct answer of which the particle's kinetic energy is at a maximum at $t = 0$.

16. A transverse wave propagates along a stretched string. Which graph below correctly shows the variation of the speed v of the wave with its frequency f?

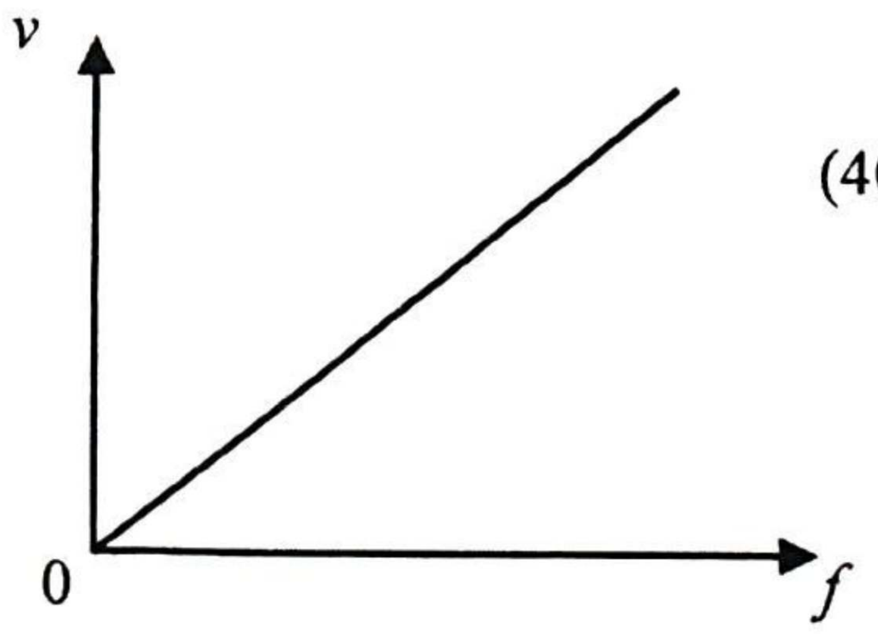
*A. (45%)



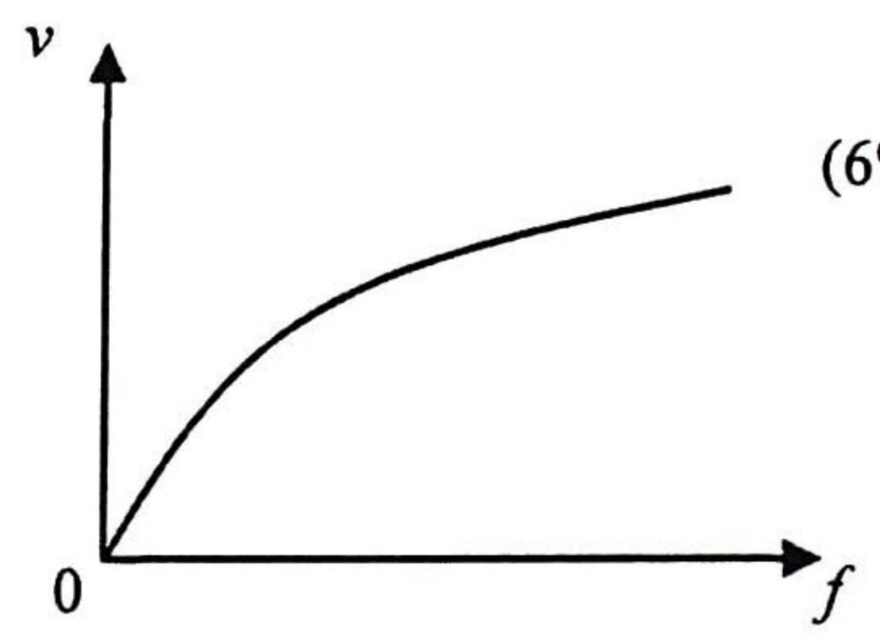
B. (9%)



C. (40%)

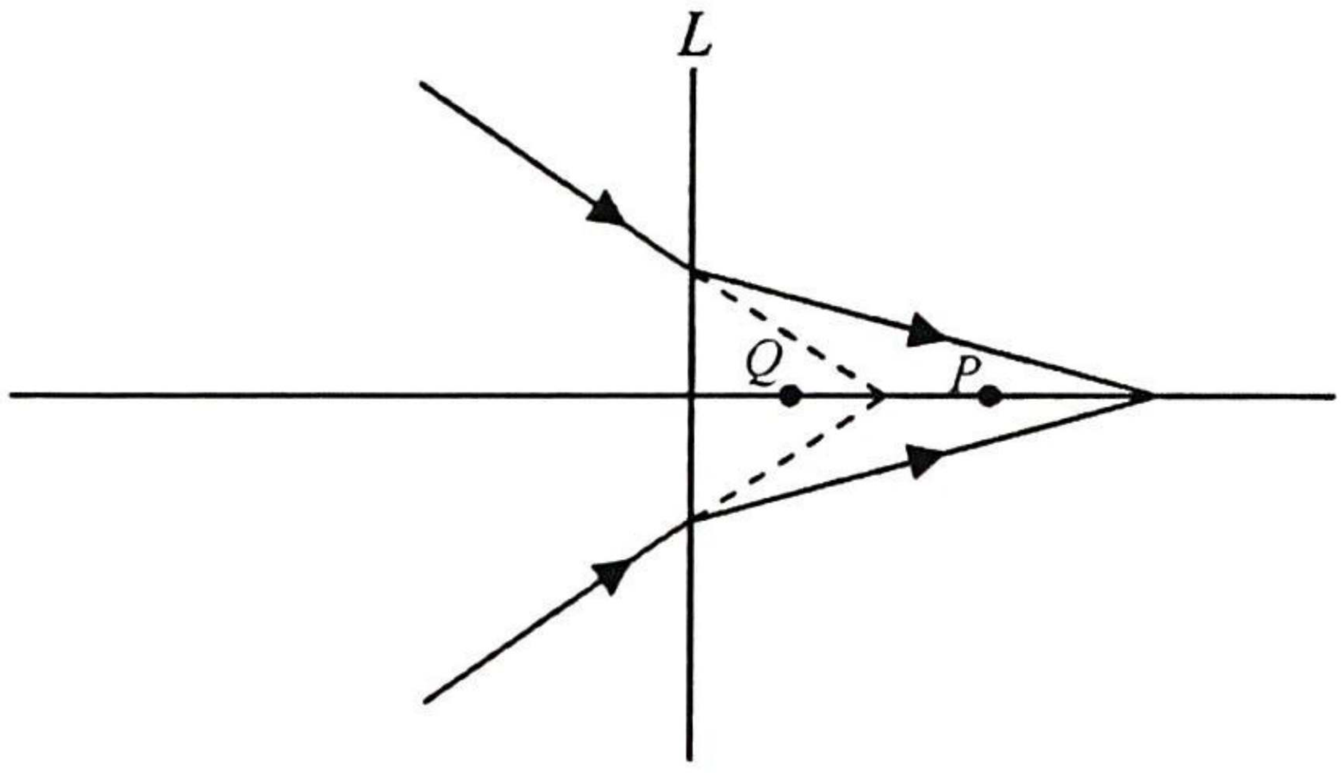


D. (6%)



40% of the candidates thought that the speed v increases linearly with frequency f and chose option C, which suggests that some candidates might have assumed the wavelength, instead of the speed, remains unchanged on a stretched string.

18.

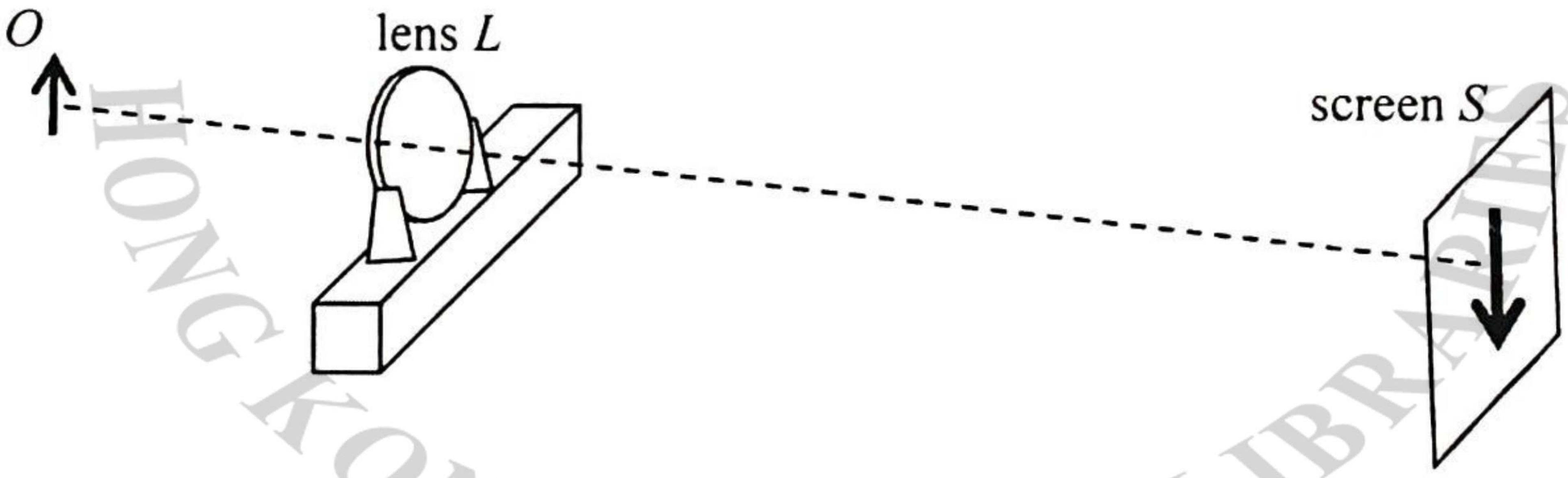


Referring to the above ray diagram, what kind of lens is represented by L ? Which point, P or Q, can be its focus ?

	lens L	focus	
*A.	concave	P	(45%)
B.	convex	P	(22%)
C.	concave	Q	(20%)
D.	convex	Q	(13%)

Candidates choosing options B and D suggests that they did not have a basic understanding of the converging and diverging properties of lenses.

20. The figure shows an enlarged sharp image of an object O formed on a screen S by a convex lens L.



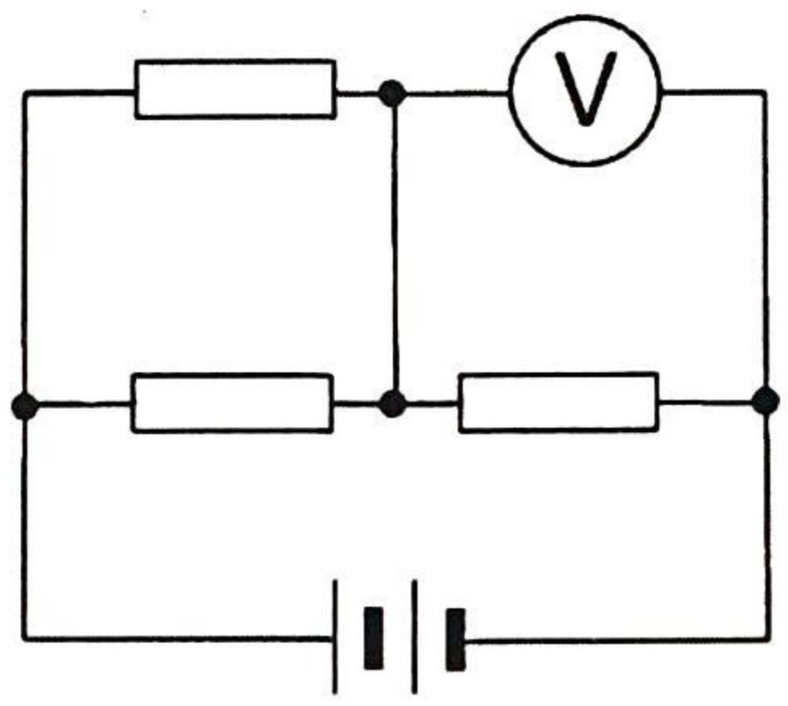
Which of the following can give a diminished sharp image on the screen ?

(1) Keeping the positions of O and L unchanged, move S suitably closer to L.
(2) Keeping the positions of L and S unchanged, move O suitably farther away from L.
(3) Keeping the positions of O and S unchanged, move L suitably closer to S.

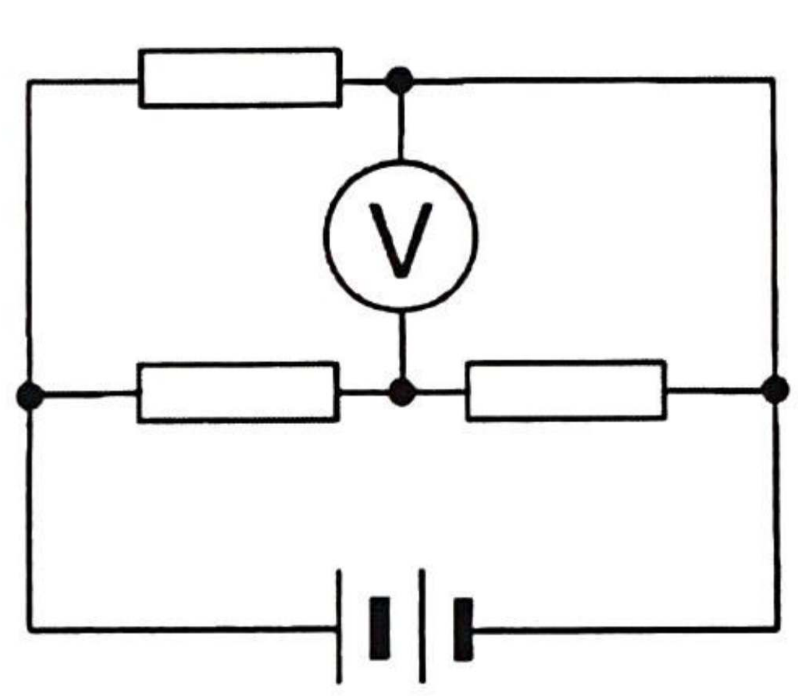
A.	(1) only	(13%)
*B.	(3) only	(31%)
C.	(1) and (2) only	(14%)
D.	(2) and (3) only	(42%)

Less than one-third of the candidates were able to answer this question correctly. It seems that many candidates were not aware that a certain object distance only corresponds to a unique image distance.

23. Three identical resistors, a battery of negligible internal resistance and an ideal voltmeter are connected to form Circuits (a) and (b) respectively.



Circuit (a)



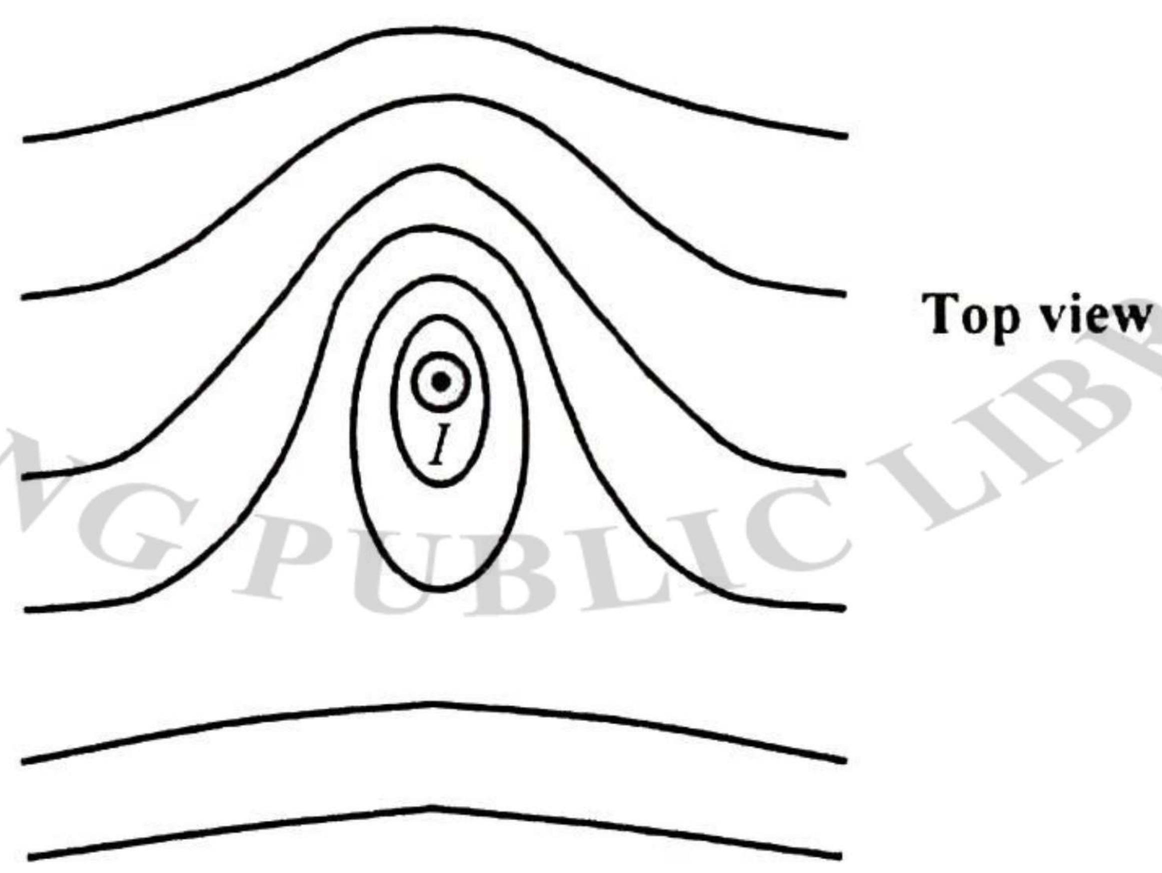
Circuit (b)

Given that the voltmeter reading is 8 V in Circuit (a), what is the voltmeter reading in Circuit (b) ?

A.	4 V	(25%)
*B.	6 V	(32%)
C.	8 V	(24%)
D.	12 V	(19%)

Just under one-third of the candidates were able to tackle the problem by treating the branch with the ideal voltmeter as an open circuit.

26. The figure below shows the magnetic field pattern on a horizontal surface around a long vertical straight wire carrying a steady current I pointing out of the paper. The Earth's magnetic field is **NOT** neglected.

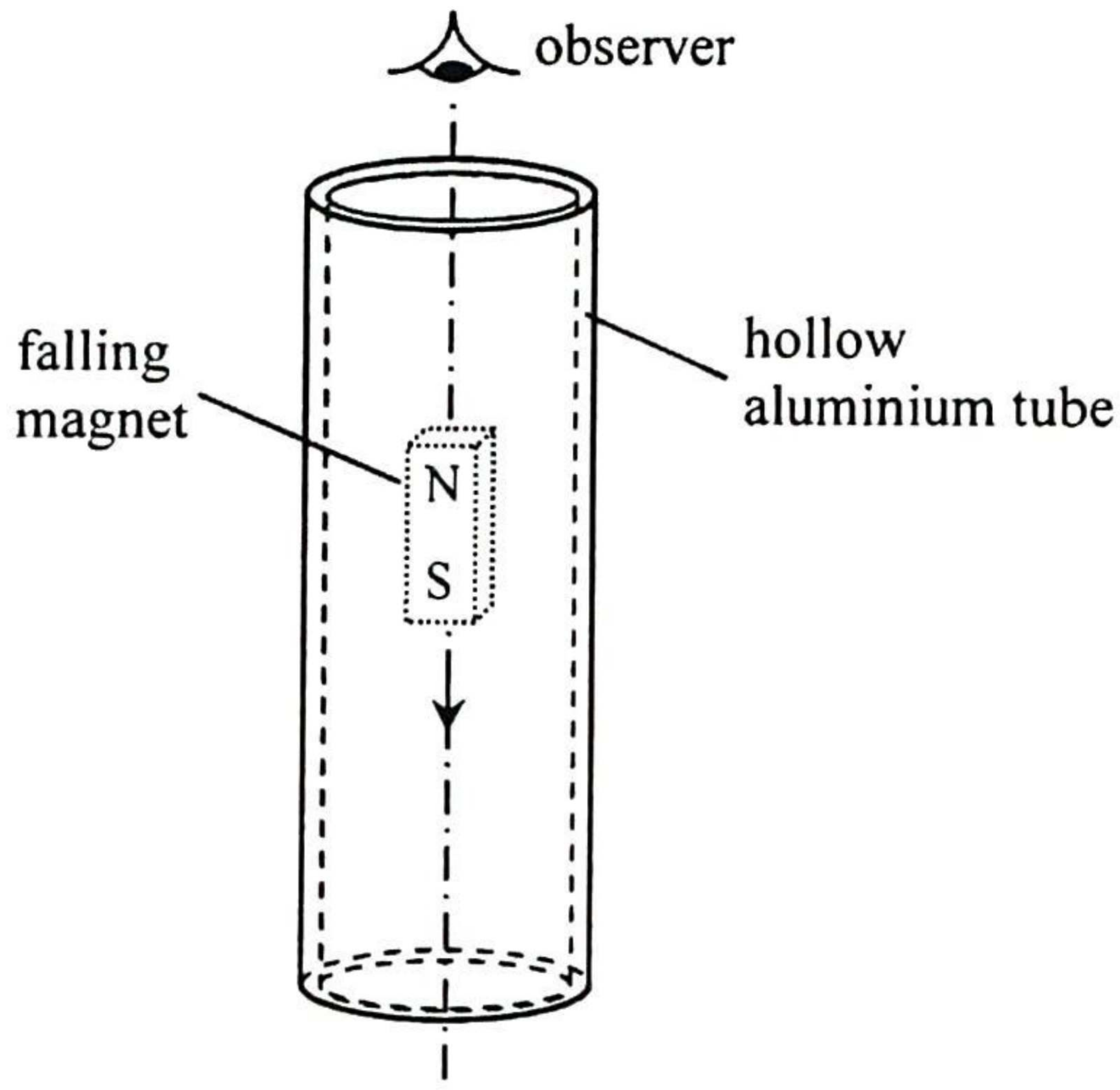


What are the directions of the following ?

	the horizontal component of the Earth's magnetic field	the magnetic force experienced by the current-carrying wire	
*A.	←	↓	(42%)
B.	←	↑	(25%)
C.	→	↓	(13%)
D.	→	↑	(20%)

Of the two-thirds of the candidates correctly identifying the direction of the magnetic field, just over 40% were also able to get the direction of magnetic force correct.

28.



When a small strong magnet falls through a hollow aluminium tube as shown, eddy currents are induced. Which of the following correctly describes the direction of current induced in the tube when viewed by an observer from above ?

A.	clockwise both above and below the magnet	(17%)
B.	anti-clockwise both above and below the magnet	(19%)
C.	clockwise above the magnet and anti-clockwise below the magnet	(32%)
*D.	anti-clockwise above the magnet and clockwise below the magnet	(32%)

Only one-third of the candidates were able to apply Lenz's law in answering this problem.

32. The decay constant of a radioisotope of an element

A.	is random.	(22%)
B.	depends on pressure and temperature.	(8%)
C.	is directly proportional to the number of nucleons in the isotope.	(21%)
*D.	is an identifying characteristic of that isotope.	(49%)

Just under half of the candidates fully understood the uniqueness of decay constant.